



MAJOR PROJECT REPORT

Phishing Awareness Simulation Using Social Engineering Techniques

1 INTRODUCTION

Phishing is one of the most prevalent cyber security threats in today's digital world. It is a form of social engineering attack where attackers impersonate trusted entities to deceive users into revealing sensitive information such as usernames, passwords, OTPs, and financial details. Unlike traditional cyber attacks that exploit software vulnerabilities, phishing primarily targets human psychology.

With the rapid growth of online services, users frequently receive emails and messages requesting urgent actions, such as password resets or account verification. Attackers take advantage of this urgency and lack of cyber awareness to compromise user accounts. Therefore, educating users and understanding their behaviour during phishing attempts is crucial for strengthening cyber security defences.

This project focuses on simulating a phishing scenario in a **controlled and ethical environment** to analyse user awareness and suggest preventive measures.

2 OBJECTIVES

The main objectives of this project are:

- To understand common phishing techniques used in real-world attacks
- To simulate a realistic phishing email in a controlled environment
- To analyse user behaviour when exposed to phishing attempts
- To measure awareness levels among users
- To suggest effective countermeasures and awareness strategies to prevent phishing attacks

METHODOLOGY

The project was conducted in a step-by-step and ethical manner as described below:

3.1 Research Phase

A detailed study was conducted on common phishing techniques such as:

- Email-based phishing
- Fake login pages
- Urgency and fear-based messages
- Spoofed sender addresses
- Trust exploitation using familiar branding

Human behavioural factors contributing to phishing success were also analysed.

3.2 Phishing Email Design (Simulation)

A fake but harmless phishing email was designed with the following characteristics:

- Urgent subject line (e.g., *Verification of account Required*)
- Professional-looking email format
- A call-to-action button leading to a fake login page

 This email was created **only for awareness purposes**, and participants were informed beforehand.

 **Screenshot Included:** Fake phishing email draft

Urgent: Account Verification Required

1 message

Biman Shil <minakshishil0@gmail.com>

To: Biman Shil <minakshishil0@gmail.com>

Dear User,

We detected unusual activity on your account.
For your security, immediate verification is required to prevent temporary suspension.

Please verify your account by clicking the link below:

 [Verify Account Now](https://sites.google.com/view/phishingawarenesssimulation)
<https://sites.google.com/view/phishingawarenesssimulation>

If you do not complete this verification within 24 hours, access to your account may be limited.

Thank you for helping us keep your account secure.

Regards,
Security Team
Account Support Services

⚠ This message is part of a cyber security awareness simulation.
Do not enter real credentials.

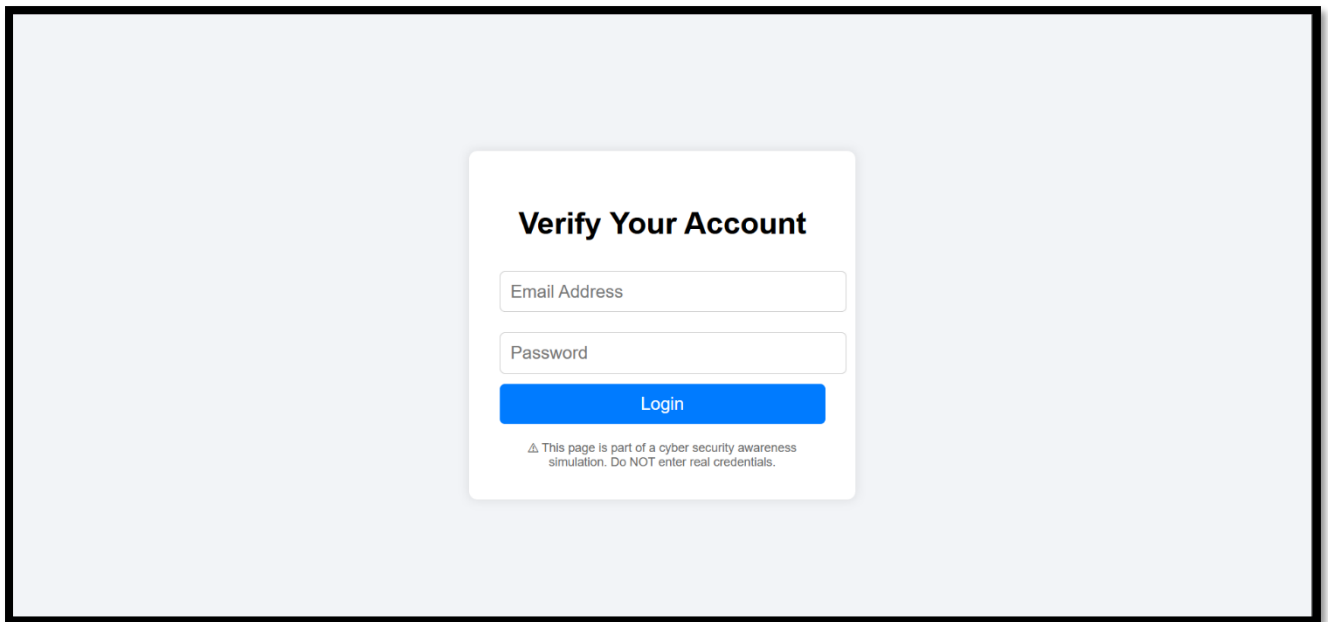
3.3 Fake Login Page Creation

A fake login page was created using **Google Forms / HTML (static)** to simulate a credential entry page.

Features:

- Fields for email/username and password
- No backend or data storage
- Clear disclaimer stating that it is a cyber awareness simulation

 **Screenshot Included:** Fake login page



3.4 Controlled Simulation

The phishing email was shared with a small group of participants (classmates/friends) who were informed that this was a cyber security awareness simulation. ***A Google Form–based awareness survey was used to collect anonymous responses and generate response summaries for analysis.***

Observed parameters:

- Users who clicked the link
- Users who attempted to enter credentials
- Users who identified or reported the email as phishing

No real credentials were collected or misused.

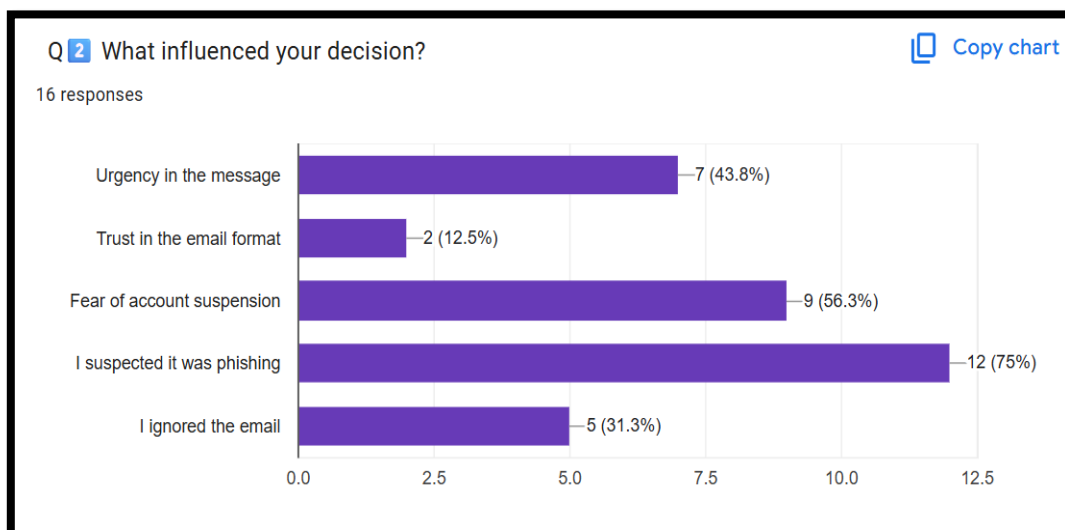
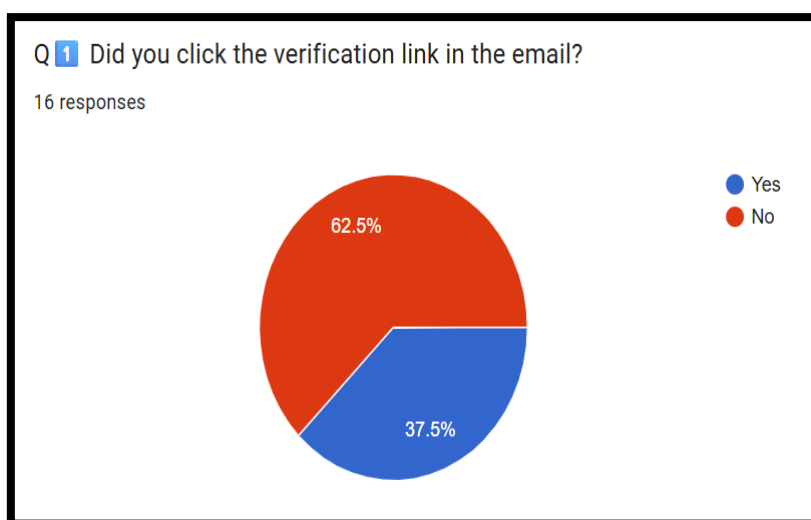
RESULTS & OBSERVATIONS

4.1 User Interaction Statistics (Sample)

Activity	Number of Users
Total participants	16

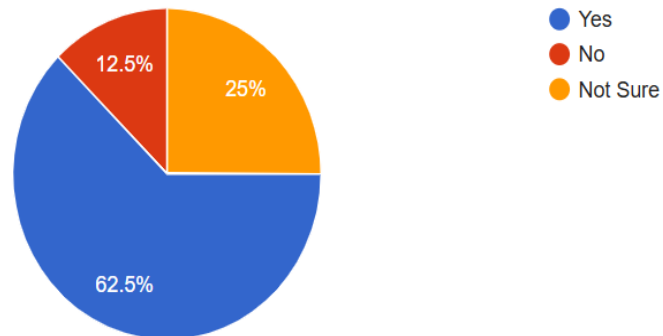
Out of the total participants, a significant number interacted with the phishing email, indicating the effectiveness of urgency-based social engineering. A smaller portion of users were able to identify the phishing attempt, highlighting the gap in cyber awareness. The results demonstrate that phishing attacks remain a high-risk threat due to human error.

 Screenshot Included: Click / response statistics



Q 3 Did you recognize the email as a phishing attempt?

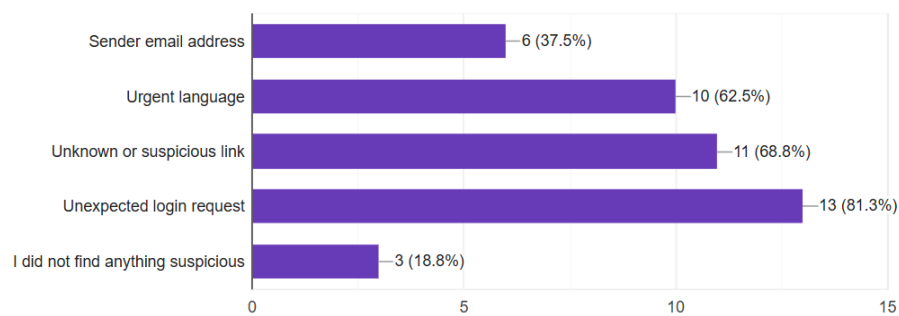
16 responses



Q 4 Which signs made the email suspicious (if any)?

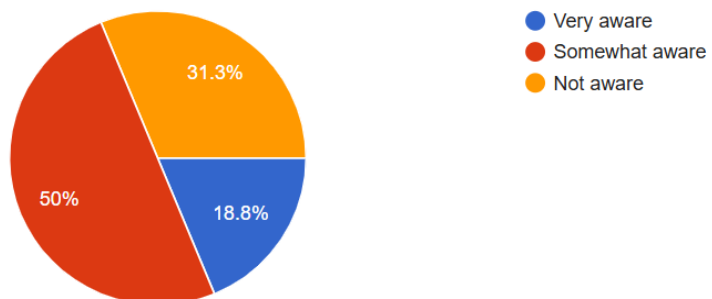
[Copy chart](#)

16 responses



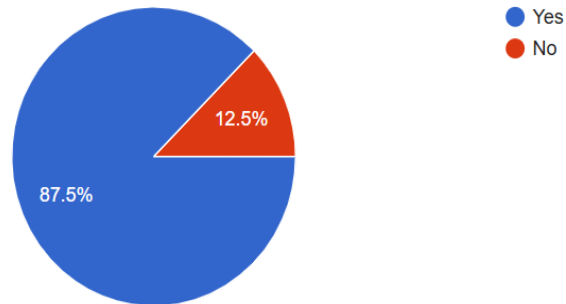
Q 5 Before this activity, how aware were you about phishing attacks?

16 responses



Q 6 After this simulation, do you feel more confident identifying phishing emails?

16 responses



Q 7 (Optional) What is one thing you learned from this simulation?

8 responses

Be safe be intelligent

Fear of data theft

Not to respond to any anonymous web

Be aware of phishing

Awareness is necessary for this type of situation to prevent data theft

Phishing is becoming common now a days

This fishing attempt should not be done

Like this simulation should be done for awareness

4.2 Behavioural Observations

- Many users clicked the link due to urgency created by the email
- Some users trusted the email format without checking the sender address
- Users with basic cyber awareness were able to identify the phishing attempt

These observations confirm that phishing attacks succeed mainly due to human error and lack of awareness rather than technical flaws.

5 LESSONS LEARNED

- Phishing attacks exploit trust, urgency, and fear
- Even technically aware users can fall victim without proper awareness
- User education is as important as technical security controls
- Ethical simulations are effective tools for improving cyber awareness

This project reinforced the importance of combining **technical knowledge with user awareness training**.

6 PREVENTIVE MEASURES & COUNTERMEASURES

To reduce phishing risks, the following measures are recommended:

6.1 User Awareness

- Regular cyber security awareness programs
- Training users to verify sender email addresses
- Encouraging users to hover over links before clicking

6.2 Technical Controls

- Email spam and phishing filters
- Multi-factor authentication (MFA)
- Secure password policies

6.3 Organizational Practices

- Conducting regular phishing simulation drills
- Clear reporting mechanisms for suspicious emails
- Cyber security policy enforcement

7 CONCLUSION/ SUMMARY

This project successfully demonstrated how phishing attacks manipulate user behaviour using social engineering techniques. Through a controlled and ethical simulation, user awareness levels were analysed, and effective preventive strategies were identified. The results highlight the need for continuous cyber awareness initiatives alongside technical security solutions to protect users from phishing threats.

This project serves as a practical demonstration of real-world cyber security challenges and defensive strategies, making it relevant for internships and entry-level cyber security roles.